

PocketSmart – FastAPI Initialisation and Import Libraries (Core Functionalities Development)

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1. Topic

Initialising the FastAPI application and importing the required libraries

Epic: Core Functionalities Development

2. Project Overview

PocketSmart is an AI-powered app that helps users track spending, set budgets, and get personalized money-saving suggestions. The FastAPI application and its libraries form the base on which all the core features of the app are built.

3. Objective

To set up the FastAPI application and import all the libraries and project modules needed, so the core functions, database, and AI model can be added on a clean and working foundation.

4. Work Steps

- 1 Create the project folder and a virtual environment for the backend.
- 2 Install the required packages, such as FastAPI, Uvicorn, SQLAlchemy, Jinja2, python-dotenv, and the Gemini SDK.
- 3 Save the installed packages in a requirements.txt file.
- 4 Create the main file and import the standard, third-party, and project libraries in a clear order.
- 5 Load the environment variables from the .env file using python-dotenv.
- 6 Initialise the FastAPI application and set its title, description, and version.
- 7 Connect the Jinja2 templates and the static files folder.
- 8 Add a simple test route to confirm the application is set up correctly.
- 9 Run the application with Uvicorn and check that it starts without import errors.

5. Main Parts of the Initialisation

- Project and environment setup
- Package installation and requirements file
- Library and module imports
- Environment variable loading

- FastAPI application initialisation
- Templates and static files connection
- Test route and server run